

A Prospective Falsification-First Architecture for Pure Genus-Two Homology-Cover Renormalization

Liang Wang¹

¹School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology,
Luoyu Road 1037, 430070, Hubei, P.R. China
wangliang.f@gmail.com

2 September 2026

Abstract

P32 studies a fixed renormalization proposal for the pure genus-two homology-cover tower $H_N = \ker(\Gamma \rightarrow H_1(\Sigma; \mathbb{Z}/N\mathbb{Z}))$. Physical lifted time is scaled by exactly $1/N$, and the logarithm of the raw lift-component product by exactly $1/N^3$. The universal endpoint asks for recovery of every oriented primitive base-owner factor. Adversarial review found that the earlier four-package presentation understated the adverse information already visible in its own prospective local factors. The present article therefore adopts a falsification-first architecture. Conditional on a future derivation of those factors, positive content $d > 1$ and zero content become the first theorem targets; they must be settled before formal or compact-uniform resources are invested in the exceptional content-one stratum. The content-one analytic program is now explicitly contingent and secondary. The project-defined inverse-limit object R_+ , its localizations and projections, and the separate zero-content Hahn object remain unresolved constructions. Likewise, any analytic tail theorem is conditional on survival of the local tests and would require an absolute logarithmic-summand majorant uniform on each frozen compact set and compatible with every specified limit order. The executed work remains literature synthesis over 26 frozen sources and review-informed argument design. No factor, mismatch, obstruction, recovery statement, panel, or limit has been proved or computed. All 26 inherited citations retain `anchor:none`; P32-S13 is bibliographically VERIFIED but remains background-only, so claim-to-passage faithfulness is INCONCLUSIVE. Arithmetic A0 remains unavailable, the formal Route-A tuple is unassigned, positive arithmetic A2 is absent, and Route B remains closed.

繁體中文摘要

本文研究純種屬二同調覆蓋塔的固定重整化方案。基底是帶標記的封閉雙曲曲面及單位速度測地流，所有者涵蓋全部同調內容的保持方向本原共軛類，逆類仍分開處理。物理提升時間嚴格乘以 $1/N$ ，原始提升分量乘積的對數嚴格乘以 $1/N^3$ ；不允許依內容或結果調整。本文採取先否認後一致的架構：只有在未來自足推導局部因子後，才先檢驗正的高內容與零內容；兩者均存活後，才可投入例外的內容一子乘積、形式拓撲與聚集上一致尾項分析。因此內容一只是附條件的次要端點，絕不能代替全所有者回復，也不能移植至另一個殘餘逆極限系統。正內容的逆極限物件、局部化與投影，以及零內容的獨立 Hahn 型物件均尚未構造。本階段僅執行封閉語料綜合與審查後方法設計；未執行所有者綁定、覆蓋計算、因子推導、係數比較、面板、尾界或極限。所有引文均無段落定位，P32-S13 的書目身分已核為 VERIFIED，但仍僅作背景來源，故逐主張支持仍屬未定；本文不宜稱阻礙、回復、新穎性或路線晉級。

Keywords: homology covers; ownerwise factors; falsification-first methods; primitive geodesics; formal products; compact-uniform convergence

中文關鍵詞：同調覆蓋；所有者局部因子；先否證方法；本原測地線；形式乘積；
緊集上一致收斂

1 Introduction, Research Question, and Contribution

Finite-cover products can agree on a selected stratum or finite panel without recovering every primitive owner. P32 fixes this distinction at the object level. The base is a marked closed hyperbolic genus-two surface with unit-speed geodesic flow and presentation

$$\Gamma = \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = 1 \rangle.$$

The cover tower is the pure homology tower $H_N = \ker(\Gamma \rightarrow H_1(\Sigma; \mathbb{Z}/N\mathbb{Z}))$. It is not the residual inverse-limit object studied in the preceding branch, and a conclusion about one tower cannot repair the other. Owners are oriented primitive surface-group conjugacy classes; inverse classes remain separately oriented. For an owner with homology vector in $\mathbb{Z}^4$, content d is the greatest common divisor of the absolute coordinates, with zero content treated as a separate case.

The normalization is immutable. Physical lifted time is multiplied by exactly $1/N$, and the logarithm of the raw lift-component product is multiplied by exactly $1/N^3$. Content-dependent changes, outcome-dependent weights, adaptive schedules, and favorable-panel selection are prohibited. The universal endpoint concerns all owner contents, not an average and not the content-one subproduct alone.

The revised research question is:

Do the prospectively derived local factors for higher-content and zero-content owners already obstruct the frozen $1/N$ and $1/N^3$ full-owner recovery scheme; and only if those adverse tests are survived, what formal and analytic obligations remain for the content-one subproduct?

The change is methodological. It does not assert that the candidate factors are correct or that an obstruction exists. Higher- and zero-content checks are earlier in the declared dependency graph because they require fewer named downstream objects than formal topology and compact-uniform tails. This is not an empirical runtime, proof-complexity, or information-gain estimate. A closed adverse result would settle the universal endpoint negatively under the fixed scheme; survival would license only the next predeclared obligation. A positive content-one statement, if eventually proved, would remain restricted.

A bounded closest-work search retained four individually named neighbors. Levitt and Vogtmann's surface-group Whitehead algorithm is closest to the finite-set owner-decision component [15]; Venkov and Zograf's finite-index Fuchsian-group factorization is closest to the cover-product component [29]; Blute, Cockett, Jacqmin, and Scott's generalized power-series construction is closest to the formal-support component [1]; and Parry and Pollicott's periodic-orbit zeta monograph is closest to the analytic-program component [21]. The verified four-component matrix is `notes/stage4_prime_closest_work_comparison_matrix_round2.json` (SHA-256 5ab85f5cc3e5d39cede06b163cf18945b1b4178b9f7d8182a9973b55074601e4).

Work	Owner algorithm	Higher/zero factor	Formal carrier	Compact-uniform program
P32-CW01	finite-set surface automorphisms; no root, bytes, inverse branch, or exhaustion	no verified overlap	no verified overlap	no verified overlap
P32-CW02	Fuchsian setting only; no owner interface	finite-index zeta factorization; no ownerwise multiplicity, zero branch, or frozen normalization	product context only; no coefficient equality or projection	analytic context only; no tower schedule or majorant
P32-CW03	no verified overlap	expansions can house candidate factors; no cover derivation	ordered-monoid/finitary-support ring; not the frozen inverse system or zero fiber	support discipline; no scalar tail theorem
P32-CW04	periodic-orbit population only; no canonicalizer	periodic-product context; no P32 content split or normalized factor	formal zeta expressions; no owner-coordinate carrier	closest analytic neighbor; no exact summand, cofinality, majorant, or interchange

Each record is nearest in only one component; none closes the joint dependency chain. This four-record search is neither exhaustive nor evidence of priority, and no source is transferred beyond its source-verified passage scope.

The word “owner” is universal in this article. It never means only a content-one class. The population comprises oriented primitive surface-group conjugacy classes across positive higher content, content one, and the separately typed zero-content case. Inverse classes remain distinct. This corrects a potentially misleading shorthand in an older state description and prevents a favorable exceptional stratum from being substituted for the frozen all-owner endpoint.

Three gates organize the program. The owner gate must bind equality, oriented conjugacy, maximal roots, inverse handling, serialization, and prefix completeness to the marked presentation. The local-factor gate must derive cover order, component multiplicity, primitive-lift status, rescaled period, and normalized exponent for each relevant content. The formal/analytic gate must supply the coefficient objects and, if local tests survive, the uniform tail theorem. A failure at an earlier gate stops later work; it is not repaired by changing a schedule, selecting a favorable panel, or moving to a different tower.

The falsification-first order is therefore informational rather than promotional. It asks the shortest exact questions capable of deciding the universal proposal while leaving survival, obstruction, and not-evaluable outcomes all open. The architecture does not infer a mismatch from the displayed shapes of unproved formulas, and it does not infer convergence from a finite diagnostic plan.

2 Frozen Literature and Theoretical Boundary

2.1 Owners, conjugacy, roots, and prefix completeness

The algorithmic corpus supplies several component surfaces. Finite combinatorial structure, word-hyperbolic conjugacy, finite-subset and finite-list decisions, and root-extraction solvability provide adjacent normal-form and decision backgrounds [3–5, 9, 16]. A recent surface-group preprint presents a possible normal-form, conjugacy, and root-finding direction, but it is not peer reviewed and its presentation has not been mapped to the frozen marked presentation [30].

These sources do not bind SG2OwnerCanonical-v1. A complete interface still requires exact equality normal form, oriented conjugacy, maximal primitive roots, separate inverse handling, deterministic serialization, and proof that a requested prefix contains every earlier eligible owner with zero unresolved decisions. Solving a supplied finite list is not the same as certifying prefix completeness.

The distinction between list decision and exhaustion is decisive. A finite-list routine starts from supplied elements; the project prefix must additionally prove that no earlier eligible owner was omitted under the frozen word-length and lexicographic order. Root extraction must be maximal, and orientation must remain external to inversion. Until these properties are proved and replayable, a panel cannot be labeled complete and a zero unresolved count cannot be inferred from a successful run on selected inputs.

2.2 Cover and homology structure

Finite abelian surface covers and Riemannian covering theory supply structural context, while homology-refined geodesic distributions and modern splitting language provide adjacent decomposition surfaces [7, 11–13, 22, 28].

This group can inform deck actions, splitting, and homology strata, but it does not prove P32’s project-specific owner order, primitive lift count, lifted period, or normalized factor. It also does not select $1/N$ or $1/N^3$. Those formulas and choices remain internal theorem targets.

In particular, structural cover vocabulary does not license the project’s displayed local factors. The later derivation must start from the exact deck action of the pure tower, determine the order of the owner image, count and type lifted components, and prove which lifts are primitive. It must then propagate both frozen normalizations without silently moving between raw products and logarithms. These are project-specific proof obligations even when neighboring cover results suggest useful tools.

2.3 Primitive-orbit products and continuation

The historical Selberg record supplies trace and zeta context only. Its bibliographic identity and metadata are VERIFIED, but this article keeps it background-only [25]. Primitive-orbit dynamical-zeta theory, prime-orbit asymptotics, and finite-cover Artin formalism supply general product and factorization contexts [2, 20, 24].

Continuation theory is represented by two routes. The first source is admitted only for the meromorphic-continuation portion left unaffected by its recorded correction; affected spectral-gap and dependent counting claims are excluded. A microlocal route supplies independent continuation context [8, 10].

These works operate at product, asymptotic, or continuation levels. They do not establish independent-owner coefficient equality under the frozen normalization, identify the project formal object with a scalar product, or justify the infinite homology-cover schedule.

2.4 Counting and tail candidates

Renewal methods, prime-geodesic growth, homology-refined orbit counting, and an exponential error-term surface provide possible tail ingredients [14, 17, 23, 26].

The required P32 theorem is stronger than the existence of a nearby asymptotic. It would need an absolute bound for the exact logarithmic summand, uniform on every frozen compact rectangle, compatible with owner strata, and sufficient for every stated iterated and diagonal limit. The frozen verification records do not establish that transfer.

2.5 Formal-topology background

Ordered division rings and well-ordered supports, an inverse limit of countably-variable polynomial rings, an unrelated use of adic completion, and an institutional power-series reference provide four formal or scalar precedents [6, 18, 19, 27].

All four sources remain background-only. Their resemblance to proposed constructions cannot define R_+ , its exponent monoid, transition maps, topology, localizations, or singleton projections. They also cannot define the separate zero-content Hahn object or validate the normalization. The frozen comparison CP-P32-004 remains unresolved.

This formal-source boundary is also a truth-condition boundary. A coefficient comparison has no fixed meaning until the coefficient ring, support order, transition maps, localization, and projection are all defined. Resemblance to a known completion or Hahn construction can motivate a definition but cannot supply it. Consequently, the article does not report a coefficient mismatch; it reports the construction and comparison tasks that would make such a statement meaningful.

3 Executed Methodology

3.1 Closed-corpus synthesis

The executed method remains a staged synthesis over frozen literature and project records. The historical Phase-2 aggregate is unchanged: 51 captured manifestations, 12 duplicates removed, 39 unique records screened, 13 excluded, and 26 sources retained. Original-session result rows and row-level decisions are unavailable and are not reconstructed. Under the separate Stage-4-prime retrieval authority, a dated metadata replay submitted all 26 exact frozen query strings to Crossref: ranks one and two for Q01–Q25 and rank one for Q26, giving exactly 51 current manifestations. All 51 rows record HTTP 200 metadata. DOI, then normalized title and year, was the declared deduplication rule. The current outcome is 19 matches to the frozen inventory, 31 out-of-scope records screened out, and one duplicate manifestation removed. This replay neither replaces the frozen 26-source corpus nor refreshes a scientific result.

The raw replay is `notes/stage4_prime_literature_replay_round2.raw.json` (SHA-256 `e9f37743af4076e3e97765026f851df69131b953a9924f914232fba3aca431e`). The complete retrieval/deduplication/screening/retention ledger is `notes/stage4_prime_literature_screening_ledger_round2.json` (SHA-256 `f8ab68dd24deb133cfecbcda7d47ef4b94036691cbe0ff70c0fa183806602c1`), with a byte-equivalent row surface in the companion TSV. Every row carries query, rank, timestamp, interface, HTTP status, candidate identity, deduplication identity, duplicate link if any, decision, frozen-inventory match, and reason. Historical and current counts remain separately labeled.

The evidence matrix retained all 26 IDs and assigned admitted and excluded claim surfaces. The synthesis grouped them into owner, cover, product, tail, and formal themes. It did not average away publication status, correction status, or background-only roles. It also preserved the distinction between source identity and theorem applicability.

The evidence synthesis was not a vote across papers. Each source-effect row recorded an admitted contribution, an excluded stronger use, and an applicability warning. Thus an algorithmic result could inform one component without being treated as the bound owner interface, and a cover theorem could provide structural context without proving the project factor. The same discipline keeps P32-S13 background-only and keeps the correction-affected claims of P32-S17 outside the admitted surface.

Citation closure is now recorded in a 30-row source-to-claim matrix at `notes/stage4_prime_claim_passage_matrix_round2.json` (SHA-256 `b4c759e35e894f80e601d12601f5d84b8635a4f78e1e9ccaacb91d93dbf87bbe`), with a companion TSV. Every row gives the component or claim role, exact passage locator or explicit null, hypotheses, correction state, applicability statement, and prohibited stronger transfer. P32-S01–P32-S26 retain `anchor:none` and `INCONCLUSIVE` theorem-passage status; their missing locators were not reconstructed. Only the narrow publisher- or author-source scopes for P32-CW01–P32-CW04 are finalized. Those four scopes position adjacent components and do not establish a P32 owner interface, factor, formal application, tail theorem, or joint novelty. No direct quotation is introduced, and metadata identity is not treated as theorem verification.

3.2 Revision procedure and nonexecution boundary

The executed scholarly method comprised corpus capture, deterministic deduplication, inclusion-frame screening, source-effect coding, thematic synthesis, and fail-closed propagation of unsupported transfers. The separately authorized Stage-4-prime supplement added a bounded exact-query metadata replay, a four-record closest-work search, and notes-side passage and artifact matrices. It performed no owner enumeration, cover calculation, factor derivation, coefficient comparison, panel, compact-uniform estimate, limit, or scientific result refresh. The following Revision-1 freeze statement governs the earlier Phase-6 drafting pass; it does not erase the later, separately recorded metadata-only supplement.

The Phase-6 ClaimIntent manifest was frozen before article prose. No new retrieval, source-finalization call, direct quotation, locator, experiment, factor computation, coefficient comparison, panel, or limit was permitted. The executed analytical operation was logical comparison of frozen definitions and review findings. Every scientific disposition remains unexecuted or not evaluated.

4 Review-Adjudicated Proof and Method Architecture

4.1 Frozen candidate factors are theorem targets

For a positive-content owner g , let its content be $d > 0$ and define prospectively $q_N(g) = \gcd(N, d)$. The frozen program asks a later proof to derive the deck order, primitive-lift count, physical lifted period, and normalized local factor, including the target expression

$$F_{N,g}(s) = (1 - \exp(-s \text{ ell}(g)/q_N(g)))^{(-q_N(g))}.$$

For zero content, the frozen program treats the owner separately and targets

$$F_{N,g^-(0)}(s) = (1 - \exp(-s \text{ ell}(g)/N))^{(-N)}.$$

Neither expression is a result of this article. A proof must derive them from the deck action under the exact marked presentation, establish the component count and covering degree, prove primitiveness of lifted components, and account for both $1/N$ time scaling and $1/N^3 \log$ multiplicity. The zero-content expression cannot be obtained by informal substitution into the positive-content case.

The base primitive-owner factor used for the proposed comparison is

$$B_g(s) = (1 - \exp(-s \text{ ell}(g)))^{(-1)}.$$

Conditional scalar lemma. If $\ell > 0$, $s > 0$ is real, and $m \geq 2$ is an integer, then

$$\Phi_m(s) = (1 - e^{-s\ell/m})^{-m} > B(s) = (1 - e^{-s\ell})^{-1}.$$

Indeed, put $x = e^{-s\ell/m}$. Then $0 < x < 1$ and $e^{-s\ell} = x^m$. Since $m \geq 2$,

$$(1 - x)^m < (1 - x) < (1 - x^m).$$

All three quantities are positive, so taking reciprocals reverses both strict inequalities:

$$(1 - x)^{-m} > (1 - x)^{-1} > (1 - x^m)^{-1}.$$

The outer terms are $\Phi_m(s)$ and $B(s)$. The proof audit is `notes/stage4_prime_conditional_scalar_lemma_audit_round2.json` (SHA-256 41759b42a0a4b329be15541a4cdb9a4f3cf4fdb30b80ecaea402d03551e5a9c1). This exact lemma concerns candidate positive real functions only; it performs no factor derivation, owner observation, coefficient execution, global obstruction, recovery determination, or Route evaluation.

The positive-content derivation contract now exposes its intermediate targets. For homology content $d > 0$ and $q_N(g) = \gcd(N, d)$, the proposed identities are deck-image order $N/q_N(g)$, lifted-component count $N^3 q_N(g)$, scaled period $\text{ell}(g)/q_N(g)$, and logarithmically normalized exponent $q_N(g)$. Chaining only those proposed fields gives the displayed $F_{N,g}$; every identity has status UNPROVED. A future ledger must bind the homology vector, content, deck action, component count, covering degree, primitive-lift proof, unscaled and scaled periods, raw exponent, both frozen normalizations, and resulting factor. The zero-content ledger remains separate, and a read-only verifier must reject any unresolved field.

The two normalizations must be visible at different levels. Multiplying physical lifted time by $1/N$ changes the exponential argument, whereas multiplying the logarithm of the raw component product by $1/N^3$ changes its exponent after component counting. Combining those operations informally can hide a missing power or an unjustified primitive-component assumption. The future proof must therefore derive the raw lift product first and apply each frozen normalization in its declared order.

4.2 First adverse target: higher content

The controlling modulus schedule is $N_k = k!$ for every integer $k \geq 1$. For a fixed positive content $d > 1$, the candidate formula would eventually have $q_N(g) = d$ whenever the factorial modulus is divisible by d . The proposed local factor would then have the stabilized displayed shape

$$(1 - \exp(-s \cdot \text{ell}(g)/d))^{(-d)}.$$

For a higher-content owner with $d \geq 2$, the lemma applies with $m = d$ only after the frozen factorial schedule reaches $d \mid N$, a valid cover calculation derives the displayed higher-content factor with $q_N(g) = d$, and the permitted finite scalar or singleton comparison is bound. Under all of those antecedents it gives the strict scalar inequality $\Phi_d(s) > B(s)$ for every real $s > 0$. None of the antecedent scientific derivations or an ownerwise execution is present here. Thus this is a conditional implication, not a reported mismatch or global obstruction; failure of any gate returns NOT_EVALUABLE.

This priority is falsification-first because it can decide the universal endpoint before a global infinite product is built. It also prevents a content-one success from concealing an adverse owner class.

The higher-content test is ownerwise and target blind. A later execution would have to select an eligible owner under the frozen owner interface, not because its displayed length or content appears favorable. It would then derive the factor and project to the independently named owner variable in the completed positive-content formal object. Only a replayable inequality under that object's exact equality relation could be labeled an obstruction. An unresolved projection or factor derivation returns a not-evaluable state instead.

Survival also has a narrow meaning. If the first registered higher-content comparison does not yield a mismatch, that result does not establish the universal scheme, the formula for every owner, or an infinite product. It licenses only the next predeclared test or proof obligation. The architecture thereby avoids turning a failed attempt at falsification into a positive recovery theorem.

4.3 Second adverse target: zero content

Zero content has order one in the prospective cover algebra, N^4 candidate lifted components, rescaled period $\text{ell}(g)/N$, and normalized exponent N . Each of those fields requires proof. Conditional on that derivation, the zero-content factor is the second local comparison target. It must be represented in the separately typed zero-content formal object and compared with the base owner factor without pretending that $d=0$ belongs to the positive-content exponent system.

For zero content and a fixed integer $N \geq 2$, the same lemma applies with $m = N$ only after the separate zero-content deck, component, primitive-lift, and normalization derivation yields its displayed factor and the one-owner H_g scalar domain is valid. Under those antecedents it gives $\Phi_N(s) > B(s)$ for every real $s > 0$. It is not obtained by substituting $d = 0$ in the positive-content system. No such derivation or owner comparison was executed, so zero-content scientific status remains NOT_EVALUABLE; the conditional inequality alone supplies no global obstruction, recovery conclusion, or Route credit.

Zero content requires this separate path because its homology image has a different algebraic status from a positive integer content. Treating $d = 0$ as a limiting or substituted positive value would obscure the component and exponent arguments. The zero-content certificate must state its own support monoid, factor semantics, and comparison map. It must also preserve the same oriented primitive-owner policy used by the other strata.

The accepted outcomes are deliberately three-valued. A closed derivation followed by equality records local survival; a closed derivation followed by certified inequality records an ownerwise adverse result for the frozen scheme; and an unclosed derivation or formal comparison records not evaluable. None of these outcomes has been produced here. In particular, the face-value appearance of the prospective exponent N is not a computed coefficient certificate.

4.4 Content one is contingent and secondary

For $d=1$, the candidate expression has $q_N=1$, making this stratum structurally exceptional within the proposed formula. That observation does not establish recovery. A content-one program still requires canonical owner enumeration, a well-defined formal product, scalar specialization, absolute compact-uniform control, cofinality, and justified limit interchange.

The branch may begin only if the higher- and zero-content theorem targets are resolved in a way that warrants further work. Even a future positive content-one result would describe a restricted subproduct. It would not reinstate universal owner recovery, repair the residual inverse-limit system, supply arithmetic A0, earn automatic positive A2 credit, or open Route B.

Content one remains scientifically interesting because the candidate formula simplifies there, but simplification is not recovery. The branch still needs a prefix-complete enumeration over oriented primitive owners, a product defined in the chosen formal topology, a justified scalar specialization, and a compact-uniform logarithmic tail. A positive statement about that subproduct must name the restriction in its title, theorem, and data surfaces. It cannot be relabeled as an all-content primitive product.

Nor can the pure tower repair the earlier residual inverse-limit object. The systems have different owners and quantified limits. Even a complete content-one theorem for the pure tower would remain a calibrator result under the fixed $1/N$ and $1/N^3$ scheme. This separation is part of the initial dynamical restriction, not a caveat to be reconsidered after outcomes.

4.5 Formal objects remain unresolved

Let $\mathcal{O}_+$ and $\mathcal{O}_0$ be disjoint typed sets of oriented primitive owners with $d(g) \geq 1$ and $d(g) = 0$, respectively, and $\ell(g) > 0$. The positive carrier retains the frozen initial system. For a finite $F \subset \mathcal{O}_+$ and $D \in \mathbb{N}_0$, set

$$\mathfrak{m}_F = (u_g : g \in F), \quad R_{F,D} = \mathbb{Q}[u_g : g \in F] / \mathfrak{m}_F^{D+1}.$$

Order indices by $(F, D) \preceq (F', D')$ exactly when $F \subseteq F'$ and $D \leq D'$. The map $\rho_{(F', D'), (F, D)} : R_{F', D'} \rightarrow R_{F, D}$ sets variables in $F' \setminus F$ to zero and then truncates total degree above D . Define

$$R_+ = \varprojlim_{(F, D)} R_{F, D}.$$

Each coordinate has the discrete topology and R_+ has the inverse-limit topology. Equality means equality in every (F, D) coordinate; convergence means eventual coordinate stability. Thus R_+ is a Hausdorff complete topological $\mathbb{Q}$ -algebra, independently of any scalar product.

Object or map	Exact typed surface	Status and remaining gate
$R_{F,D}, \rho, R_+$	finite-owner/degree quotients; inverse-limit topology and coordinate equality	DEFINED; carrier compatibility proved
A_F, j_F	declared unit denominators and embedding into R_+	DEFINED; factor derivation pending
$R_0 = \bigsqcup_g \{g\} \times H_g$	separately typed one-owner rational-exponent Hahn fibers	DEFINED; no multivariate zero product
π_g	$R_+ \rightarrow \mathbb{Q}[[u_g]]$, compatible at every degree	DEFINED; no global-product conclusion
finite scalar maps	$A_F \rightarrow \mathbb{C}$ and stated zero-fiber subalgebra to $\mathbb{C}$, $\operatorname{Re} s > 0$	DEFINED; no map on all R_+ or H_g
factor application	project factor to carrier and permitted owner comparison	NOT_EVALUABLE; cover derivations absent
majorant and AN-1–AN-5	exact logarithmic summands on each frozen compact	UNPROVED; enumeration, tail, and interchange absent

Here DEFINED refers only to the self-contained algebraic carrier. The broader phrases “formal object unresolved” or “not closed” elsewhere retain their scientific meaning: no project factor has been derived into the carrier, no global product or infinite scalar specialization exists, and no recovery or Route disposition changes.

For finite $F \subset \mathcal{O}_+$, define the exact localization

$$A_F = \mathbb{Q}[u_g : g \in F][(1 - u_g^r)^{-1} : g \in F, r \in \mathbb{N}].$$

Every denominator has constant term one. Its truncated geometric inverse in each $R_{E,D}$ defines a canonical homomorphism $j_F : A_F \rightarrow R_+$, setting variables in $F \setminus E$ to zero at coordinate (E, D) . For $\text{Re } s > 0$, the finite scalar map $\sigma_{s,F} : A_F \rightarrow \mathbb{C}$ is $u_g \mapsto e^{-s\ell(g)/d(g)}$; all denominator images are nonzero. No map from all of R_+ is declared.

For each $g \in \mathcal{O}_0$, let

$$H_g = \left\{ \sum_{q \in S} a_q z_g^q : a_q \in \mathbb{Q}, S \subset \mathbb{Q}_{\geq 0} \text{ well ordered} \right\}.$$

Equality is rational-exponent coefficientwise and the valuation topology has neighborhoods $v(f) > r$, where $v(f) = \min \text{supp } f$. The zero type is the tagged family $R_0 = \bigsqcup_{g \in \mathcal{O}_0} \{g\} \times H_g$: operations occur only inside one H_g , with no cross-owner product and no coercion to R_+ . For fixed N , put $A_{g,N}^0 = \mathbb{Q}[z_g^{1/N}][(1 - z_g^{1/N})^{-1}, (1 - z_g)^{-1}] \subset H_g$ and define $\tau_{s,g,N} : A_{g,N}^0 \rightarrow \mathbb{C}$ by $z_g^q \mapsto e^{-s\ell(g)q}$ for $\text{Re } s > 0$. The literature comparison CP-P32-004 remains unresolved and is not a premise of these definitions.

Formal compatibility lemma. The transition maps above are well-defined continuous unital homomorphisms, satisfy identity and composition, and make R_+ Hausdorff and complete. The maps j_F are injective and commute with finite-owner restriction. For every $g \in \mathcal{O}_+$,

$$\pi_g : R_+ \longrightarrow \mathbb{Q}[[u_g]] = \varprojlim_D \mathbb{Q}[u_g]/(u_g^{D+1})$$

is a well-defined continuous unital homomorphism satisfying $\pi_{g,D} = \rho_{(F,D),(\{g\},D)} \circ p_{F,D}$ for every finite $F \ni g$. Hahn convolution and the stated finite scalar maps are likewise well defined.

Proof. Variable deletion preserves total degree; hence an element of $\mathfrak{m}_{F'}^{D'+1}$ maps to zero modulo $\mathfrak{m}_F^{D+1}$ when $D \leq D'$. Substitution and truncation compose, proving the inverse-system laws. The limit is the closed equalizer of these laws inside a product of complete Hausdorff discrete algebras, so it is complete and Hausdorff. In every truncation, $(1 - u_g^r)^{-1}$ is the finite geometric sum through degree D , and these sums commute with transition. If $j_F(p/q) = 0$, multiplying by the unit expansion of q gives $p = 0$ in every degree, hence $p = 0$; thus j_F is injective. The inverse-limit equations make π_g independent of F and prove its displayed compatibility. For well-ordered supports in $\mathbb{Q}_{\geq 0}$, sums of supports are well ordered and each exponent has finitely many decompositions: an infinite family would contain increasing exponents in one support and force a descending family in the other. Hahn convolution is therefore coefficient-finite; the candidate supports $\{r/N : r \geq 0\}$ and $\{r : r \geq 0\}$ qualify. Finally, $\text{Re } s > 0$ makes every evaluated monomial have modulus below one, so the declared denominator images are nonzero. The full audit is `notes/stage4_prime_formal_definition_audit_round2.json` (SHA-256 843e2119d1bddf4e63a73e58e1e13bee8520eb30cf45b608076b85967682a125).

Singleton projection supplies a one-way falsification rule. If two elements of R_+ are equal, their π_g images are equal; therefore a proved inequality in one compatible singleton coordinate disproves equality of those two already-defined elements. The same-owner zero-content comparison is made only inside its tagged H_g fiber. Conversely, singleton projections are not jointly faithful on mixed-owner monomials: for distinct g, h , every singleton projection kills $u_g u_h$. Thus the lemma neither constructs nor recovers a global product.

The conditional scalar inequality can witness a local difference only after the corresponding project factor is independently derived and mapped through the declared finite scalar domain. Those scientific antecedents remain absent, so no ownerwise observation, obstruction, recovery, or Route credit follows.

4.6 Analytic tail work is survival-conditional

Only a surviving content-one formal branch reaches the analytic stage. The scalar theorem target would need to state exact functions, summation order, owner exhaustion, modulus schedule, compact domain, and limit order. The frozen compact rectangles are

$$K(\delta, T, R) = \{s : 1 + \delta \leq \operatorname{Re}(s) \leq R, \quad |\operatorname{Im}(s)| \leq T\},$$

for every $\delta > 0$, finite $T \geq 0$, and finite $R \geq 1 + \delta$. The schedules remain

$$N_k = k!, \quad m_k = 2^k, \quad N'_k = 2 \cdot (k!), \quad k \geq 1.$$

Let $g_1, g_2, \dots$ denote a prospective certified cofinal order of content-one owners, and put $q_N(g_j) = 1$. Conditional on the factor derivation, define the exact logarithmic summand by

$$a_{j,N}(s) = -q_N(g_j) \operatorname{Log}_0 \left(1 - e^{-s\ell(g_j)/q_N(g_j)} \right) = q_N(g_j) \sum_{r \geq 1} \frac{e^{-rs\ell(g_j)/q_N(g_j)}}{r}, \quad L(m, N; s) = \sum_{j=1}^m a_{j,N}(s).$$

Here Log_0 is defined by the displayed absolutely convergent series and tends to zero as $\operatorname{Re} s \rightarrow \infty$. Every row below uses $K = K(\delta, T, R)$ with $\delta > 0$, finite $T \geq 0$, and finite $R \geq 1 + \delta$, and one of the separate schedules $S_k \in \{k!, 2(k!)\}$. The explicit pointwise bound is

$$M_{j,N,K} = -q_N(g_j) \log \left(1 - e^{-(1+\delta)\ell(g_j)/q_N(g_j)} \right),$$

which becomes $M_{j,K} = -\log(1 - e^{-(1+\delta)\ell(g_j)})$ for content one.

ID	Indices and coupling	Limit on K and exact interchange	Majorant obligation	Prerequisites/status
AN-1	fix schedule and k ; $N = S_k$; $m \rightarrow \infty$	$\lim_m L(m, S_k) = \sum_{j \geq 1} a_{j,S_k}$, uniformly on K	prove $\sum_j M_{j,S_k,K} < \infty$	factor, branch, cofinal owner order; UNPROVED
AN-2	fix m ; $k \rightarrow \infty$, separately for both S_k	$\lim_k \sum_{j \leq m} a_{j,S_k} = \sum_{j \leq m} \lim_k a_{j,S_k}$ on K	for $j \leq m$, $\sup_{k,s \in K} a_{j,S_k}(s) \leq M_{j,K} < \infty$	factor and content-one applicability; UNPROVED
AN-3	first $m \rightarrow \infty$, then $k \rightarrow \infty$	$\lim_k \sum_j a_{j,S_k} = \sum_j \lim_k a_{j,S_k}$, uniformly on K	one k -independent $\sum_j M_j(K) < \infty$	AN-1, component limits, cofinality; UNPROVED
AN-4	first $k \rightarrow \infty$, then $m \rightarrow \infty$	justify $\lim_m \lim_k L = \lim_k \lim_m L$ and the common owner sum on K	same summable $M_j(K)$ and component convergence	AN-2/AN-3, common order; UNPROVED
AN-5	$m_k = 2^k$, $N = S_k$, both schedules	identify $\lim_k L(2^k, S_k)$ with the common AN-3/AN-4 limit on K	$\sum_j M_j(K) < \infty$ and $\sum_{j > 2^k} M_j(K) \rightarrow 0$	cofinal diagonal and common iterated limit; UNPROVED

The machine-readable registry audit is `notes/stage4_prime_analytic_registry_audit_round2.json` (SHA-256 8dea4c48f10b8649dae90a1b86c05e2ca2dd67d6c526cc04a48d3214f6b9ddd0).

The five rows are obligations, not limit results. The prospective content-one identity $q_N = 1$ makes the displayed candidate summand independent of N , but it cannot replace the missing factor derivation, certified cofinal owner order, or absolute owner tail. AN-1 fixes k ; AN-2 fixes m ; AN-3 and AN-4 declare opposite iterated orders; and AN-5 couples $m_k = 2^k$ to each full infinite modulus schedule. A proof for one row or one schedule cannot be transferred silently to another. The prospective $1 \leq k \leq 8$ prefix and

panels $m \in \{8, 16, 32, 64, 128\}$ remain unexecuted diagnostics and have no convergence, cofinality, majorant, interchange, infinite-scalar, or scientific force.

The factorial and neighboring schedules act as fixed stress paths. They may not be shortened after an inconvenient content appears, and the finite prefix through $k = 8$ remains only a prospective replay and serialization diagnostic. No observation on that prefix, even exact, would establish cofinality, convergence, analytic continuation, or a limit identity. This keeps numerical or finite symbolic checks in their proper evidence class.

5 Evidence-Synthesis Findings

Eight frozen positions govern the article. First, the algorithmic corpus supplies separate normal-form, conjugacy, root, list-decision, and serialization candidates but does not bind SG2OwnerCanonical-v1. Second, higher-content and zero-content formulas are adverse local theorem targets conditional on a future self-contained derivation. Third, the program is falsification-first: those targets precede formal or compact-uniform investment in content one.

Fourth, a content-one subproduct is contingent and secondary. Fifth, R_+ , its localizations and projections, and the separate zero-content object remain construction obligations because CP-P32-004 is unresolved. Sixth, only after local survival would the analytic branch require compact-uniform absolute tail control compatible with every frozen limit order. Seventh, the exact schedules remain unchanged and finite prefixes remain unexecuted diagnostics. Eighth, the revision concerns only the pure tower and changes no scientific recovery or Route status.

These are literature-synthesis and research-design findings. They do not constitute factor derivations, coefficient certificates, obstruction theorems, convergence results, or novelty findings.

6 Reproducibility and Prospective Interface

Every artifact claimed current in this section is enumerated in notes/stage4_prime_reader_artifact_manifest_round2.json (SHA-256 16f08bccd16d70e2d4403b0f4a2e59d1e54e7e0e09a5987a229bf519ca00b3ef; 17,968 bytes; schema round10-stage4-prime-reader-artifact-manifest/1.0). Its stable public base is the exact repository commit d29a829b4acac29ff8429724467409e9820a8fa2 at https://github.com/maris205/hilbert-polya-structure/tree/d29a829b4acac29ff8429724467409e9820a8fa2/flow_systems/papers/32-homology-cover-renormalization-uniformity. The manifest gives path, full SHA-256, byte count, schema version or explicit non-schema media type, access state, and bounded evidentiary role for all 11 claimed Section-6 inputs: source inventory, verification rows, Phase-4 report, four Phase-5 reviews, synthesis, Phase-6 checkpoint, frozen methodology contract, and ClaimIntent manifest. All 11 were byte-verified at that commit.

The same manifest inventories 14 current Stage-4-prime notes sidecars as LOCAL_NOTES_SIDE CAR_NOT_PRESENT_AT_PINNED_COMMIT. They are not misrepresented as remotely reader-resolved; repository synchronization and post-apply audit remain pending. The commit locator is content-addressed but is not a persistent archive, release DOI, or preservation guarantee. Artifact hash closure does not make prose generation deterministic, passage transfer theorem-verified, or any scientific result reproducible.

A later scientific interface should freeze producer and verifier roles before outcomes. The owner layer must bind the exact marked presentation, equality, oriented conjugacy, maximal roots, inverse policy, serialization, and prefix completeness. The cover layer must bind deck action, component count, primitive-lift proof, physical period, and normalization. The formal layer must bind the positive and zero-content objects separately. Only after those bindings pass should the adverse singleton comparisons run.

The smallest next package is therefore not an owner panel or numerical grid. It is a proof-binding dossier for one higher-content owner and one zero-content owner selected without target inspection. The dossier should expose every algebraic premise and return a typed result: derivation closed and comparison eligible, derivation failed and not evaluable, or later certified mismatch/survival after separately authorized scientific execution.

That dossier should be reproducible without access to hidden producer state. It should include the frozen input and presentation hashes, the owner certificate, homology vector and content, theorem and schema versions, deck-action derivation, primitive-lift evidence, normalization trace, formal-object version, projection proof, verifier receipt, and a typed disposition. Positive and negative cases should use the same field contract. Missing evidence should invalidate the comparison rather than be represented by a default value.

Only after the two local dossiers pass their interface checks would a population manifest be meaningful. Such a manifest would freeze the target-blind enumeration rule and exact schedules before any factors are read. This separation prevents implementation debugging, owner selection, and scientific interpretation from occurring in one outcome-dependent loop. None of these dossiers or manifests exists in the present manuscript.

7 Discussion and Implications

The falsification-first order changes the value proposition of P32. The earlier design placed owner binding, lift algebra, formal construction, and analytic tails in one broad progression. The revised architecture recognizes that not every package is needed to test the universal claim. If higher or zero content fails under an exact local-factor theorem, the full-owner endpoint is settled for the frozen normalization before the content-one tail problem becomes relevant.

This does not prejudge the outcome. The candidate factors are themselves unproved, and the formal truth conditions are unfinished. The method therefore has two early stopping surfaces: failure to derive the factor, or a later certified adverse comparison. Both are scientifically clearer than continuing to an infinite product while the local ingredients remain unresolved.

Content one still has potential value as a restricted analytic object. Its exceptional status must remain visible. A favorable theorem there would not erase an ownerwise obstruction elsewhere, and it would not create arithmetic specificity in a generic homology-cover calibrator. The pure tower has no intrinsic rational-prime owner map or $\log p$ clock in this record.

The Route boundary follows directly. A0 is unavailable, so P32 is not a primary Hilbert–Pólya dynamics candidate. No stable dynamical determinant or divisor comparison has been produced, A2–A4 remain unrun, and no Route-B entry condition exists. A negative local theorem could still be a meaningful calibration result, but it would not be Route promotion.

8 Acknowledged Limitations

The dated supplement exposes all 51 current replay manifestations and their row decisions, but it is not a reconstruction of the unavailable historical 51 rows. Its 19 inventory matches, 31 out-of-scope screens, and one duplicate do not replace or recalculate the frozen 26-source corpus. The 30-row claim-passage matrix preserves INCONCLUSIVE status for all 26 inherited `anchor:none` uses and finalizes only four narrow closest-work method scopes. Exact theorem passages and hypotheses for the inherited uses remain unavailable; general correction, retraction, source-conflict, and conflict-of-interest screening was not expanded. The new sidecars are local and pending repository synchronization. No missing evidence or remote access state is reconstructed.

P32-S13 is bibliographically VERIFIED but remains background-only. P32-S06 remains a non-peer-reviewed preprint whose presentation mapping is unproved. P32-S17 remains restricted to the correction-unaffected continuation surface. The formal cluster remains background-only. General retraction and source-conflict/COI checks remain NOT_CHECKED and UNKNOWN_NOT_CHECKED.

The core mathematical objects are unresolved. SG20wnerCanonical-v1 is unbound. The positive- and zero-content factors are derivation targets. R_+ , its maps and projections, and the separate zero-content object are not closed constructions. No compact-uniform majorant or limit theorem exists in the project artifacts. No diagnostic was run.

The corpus and article also do not establish novelty. The output is deliberately positioned as a bounded research-program study. AI-assisted roles used one Codex model family with procedural separation rather than statistically independent validation. Human gate approval does not substitute for source-passage reading or mathematical proof.

9 Future Work

Future work should preserve the falsification-first order:

1. perform a separately authorized source-finalization pass for exact theorem passages, hypotheses, correction state, and any required retraction/source-conflict checks;
2. bind SG20wnerCanonical-v1 to the exact marked presentation with an independent verifier and prefix-completeness theorem;
3. derive the higher-content and zero-content owner factors from the deck action, including primitive-lift status and both fixed normalizations;
4. construct the positive-content and zero-content formal objects separately and prove singleton projection validity;
5. execute one target-blind higher-content and one zero-content comparison only under new scientific authorization;
6. stop the universal program upon a certified adverse result, or document survival without promoting it to recovery; and
7. only after survival, formulate and prove the content-one compact-uniform tail theorem on every frozen schedule and limit order.

The finite diagnostic prefix may later test serialization and replay. It cannot establish a limit and should not be scheduled before the relevant proof surfaces close.

10 Conclusion

P32 now places its shortest discriminating obligations first. Higher-content and zero-content owner factors are the primary conditional theorem targets. The content-one formal and analytic program is secondary and may proceed only after the adverse local tests survive. The formal objects remain unresolved, and tail work remains conditional on both local survival and a self-contained scalar theorem.

This manuscript is a design-level advance, not a mathematical obstruction or recovery result. No owner interface, factor derivation, coefficient test, panel, compact-uniform bound, limit, or scientific computation was executed. All citation passages remain unresolved, P32-S13 is bibliographically VERIFIED but remains background-only, arithmetic A0 remains unavailable, the formal Route-A tuple remains UNASSIGNED, positive arithmetic A2 remains absent, and Route B remains closed.

Declarations

Author contribution and accountability. Liang Wang specified the inherited mathematical object and its restrictions, supplied the conceptual direction, approved the recorded stage gates, adjudicated the falsification-first design, and is responsible for the manuscript’s scholarly content. AI assistance is disclosed below and is not credited with authorship. Stage-gate approval does not attest that the author personally performed full-text or exact source-passage verification.

Funding. No funding was received for this work.

Competing interests. The author declares no competing interests.

Ethics statement. Human-subjects and animal-research review is not applicable. The work concerns theoretical mathematics, published-source records, and project-owned workflow artifacts; it involves no participants, identifiable personal data, animals, recruitment, or intervention.

Data and materials availability. The exact inventory is `notes/stage4_prime_reader_artifact_manifest_round2.json` (SHA-256 `16f08bccd16d70e2d4403b0f4a2e59d1e54e7e0e09a5987a229bf519ca00b3ef`; 17,968 bytes). It records a schema or explicit media type, byte count, full digest, access state, and bounded role for every listed file. The 11 artifacts claimed current in Section 6 are byte-identical at public commit `d29a829b4acac29ff8429724467409e9820a8fa2`; its resolving base is https://github.com/maris205/hilbert-polya-structure/tree/d29a829b4acac29ff8429724467409e9820a8fa2/flow_systems/papers/32-homology-cover-renormalization-uniformity. The 14 Stage-4-prime support files, including the versioned notes-side bibliography, replay ledger, passage matrix, and proof audits, are explicitly marked local and absent from that pinned commit pending synchronization. The canonical bibliography, manuscript, PDF, code, experiments, results, routes, and README remain unchanged. The commit is not claimed to be a persistent archive or DOI. No owner panel, cover computation, coefficient table, tail dataset, factor certificate, or scientific result was generated.

AI Disclosure and Verification Limitation

OpenAI Codex, using the GPT-5 model family, assisted during the session dated 2026-09-02 UTC; the exact backend snapshot/build was not exposed. Recorded AI-assisted work included literature-search support, source-identity and metadata checking, evidence-matrix construction, evidence synthesis, report drafting, four role-based Phase-5 reviews, review synthesis, ClaimIntent-constrained Revision-1 drafting, citation-ID/reference closure checks, source-status preservation checks, and stable-ID revision accounting. No AI system executed an owner interface, cover calculation, factor derivation, coefficient test, numerical panel, limit proof, experiment, obstruction certificate, or canonical-results refresh.

Liang Wang is the responsible human author. He approved the project restrictions, workflow gates, and the Phase-6 falsification-first design choice. Those approvals do not state that he personally read every source in full or verified any claim at the exact source-passage level. The recorded source work was bounded mainly to identity, metadata, abstracts, authoritative landing pages, correction records, and project-local claim-fitness judgments. All 26 citations lack passage locators; claim-to-passage faithfulness remains INCONCLUSIVE. The article does not claim theorem-level source verification, novelty, impossibility, or a clean retraction/conflict screen.

Development provenance. Four procedurally separated review labels—citation integrity, Devil’s Advocate, editorial, and ethics/integrity— were instantiated within one Codex model family. Their same-family, correlated-error limitation remains explicit; role separation is not independent validation. The recorded MAJOR_REVISION code and Liang Wang’s author adjudication are workflow history, not evidence for a factor, formal equality, inequality application, obstruction, recovery result, or limit theorem. The 26 inherited uses in the preceding paragraph still lack passage locators; the four new closest-work records have only their separately bounded narrow scopes.

References

- [1] Richard Blute, Robin Cockett, Pierre-Alain Jacqmin, and Philip Scott. Finiteness spaces and generalized power series. *Electronic Notes in Theoretical Computer Science*, 341:5–22, 2018. doi: 10.1016/j.entcs.2018.11.002. URL <https://doi.org/10.1016/j.entcs.2018.11.002>.
- [2] Eliot Brenner and Florin Spinu. Artin formalism for Selberg zeta functions of co-finite Kleinian groups. *Journal de Théorie des Nombres de Bordeaux*, 21(1):59–75, 2009. doi: 10.5802/jtnb.657.
- [3] Martin R. Bridson and James Howie. Conjugacy of finite subsets in hyperbolic groups. *International Journal of Algebra and Computation*, 15(4):725–756, 2005. doi: 10.1142/S0218196705002529.
- [4] David J. Buckley and Derek F. Holt. The conjugacy problem in hyperbolic groups for finite lists of group elements. *International Journal of Algebra and Computation*, 23(5):1127–1150, 2013. doi: 10.1142/S0218196713500203.
- [5] James W. Cannon. The combinatorial structure of cocompact discrete hyperbolic groups. *Geometriae Dedicata*, 16(2):123–148, 1984. doi: 10.1007/BF00146825.
- [6] Christopher Chiu, Tommaso de Fernex, and Roi Docampo. Embedding codimension of the space of arcs. *Forum of Mathematics, Pi*, 10:e4, 2022. doi: 10.1017/FMP.2021.19.
- [7] Gunther Cornelissen and Norbert Peyerimhoff. Homological wideness, “class field theory” for covers, and a number theoretical analogue. In *Twisted Isospectrality, Homological Wideness and Isometry*, pages 85–90. Springer, 2023. doi: 10.1007/978-3-031-27704-7_10.
- [8] Semyon Dyatlov and Maciej Zworski. Dynamical zeta functions for Anosov flows via microlocal analysis. *Annales Scientifiques de l’École Normale Supérieure*, 49(3):543–577, 2016. doi: 10.24033/ASENS.2290.
- [9] David B. A. Epstein and Derek F. Holt. The linearity of the conjugacy problem in word-hyperbolic groups. *International Journal of Algebra and Computation*, 16(2):287–305, 2006. doi: 10.1142/S0218196706002986.
- [10] Paolo Giulietti, Carlangelo Liverani, and Mark Pollicott. Anosov flows and dynamical zeta functions. *Annals of Mathematics*, 178(2):687–773, 2013. doi: 10.4007/annals.2013.178.2.6.
- [11] S. A. Jassim. Finite abelian surface coverings. *Glasgow Mathematical Journal*, 25(2):207–218, 1984. doi: 10.1017/S0017089500005632.
- [12] Atsushi Katsuda and Toshikazu Sunada. Homology and closed geodesics in a compact riemann surface. *American Journal of Mathematics*, 110(1):145–155, 1988. doi: 10.2307/2374542.

- [13] Atsushi Katsuda and Toshikazu Sunada. Closed orbits in homology classes. *Publications Mathématiques de l’IHÉS*, 71:5–32, 1990. doi: 10.1007/BF02699875.
- [14] Steven P. Lalley. Renewal theorems in symbolic dynamics, with applications to geodesic flows, noneuclidean tessellations and their fractal limits. *Acta Mathematica*, 163:1–55, 1989. doi: 10.1007/BF02392732.
- [15] Gilbert Levitt and Karen Vogtmann. A whitehead algorithm for surface groups. *Topology*, 39(6):1239–1251, 2000. doi: 10.1016/S0040-9383(99)00027-0. URL [https://doi.org/10.1016/S0040-9383\(99\)00027-0](https://doi.org/10.1016/S0040-9383(99)00027-0).
- [16] I. G. Lysenok. On some algorithmic properties of hyperbolic groups. *Mathematics of the USSR-Izvestiya*, 35(1):145–163, 1990. doi: 10.1070/IM1990v035n01ABEH000693.
- [17] G. A. Margulis. Applications of ergodic theory to the investigation of manifolds of negative curvature. *Functional Analysis and Its Applications*, 3(4):335–336, 1969. doi: 10.1007/BF01076325.
- [18] B. H. Neumann. On ordered division rings. *Transactions of the American Mathematical Society*, 66(1):202–252, 1949. doi: 10.1090/S0002-9947-1949-0032593-5.
- [19] NIST Digital Library of Mathematical Functions Project. DLMF section 4.6: Power series. NIST Digital Library of Mathematical Functions, 2026. URL <https://dlmf.nist.gov/4.6>.
- [20] William Parry and Mark Pollicott. An analogue of the prime number theorem for closed orbits of Axiom A flows. *Annals of Mathematics*, 118(3):573–591, 1983. doi: 10.2307/2006982.
- [21] William Parry and Mark Pollicott. *Zeta Functions and the Periodic Orbit Structure of Hyperbolic Dynamics*. Number 187–188 in Astérisque. Société mathématique de France, 1990. URL https://numdam.org/item/AST_1990__187-188__1_0/.
- [22] Ralph Phillips and Peter Sarnak. Geodesics in homology classes. *Duke Mathematical Journal*, 55(2):287–297, 1987. doi: 10.1215/S0012-7094-87-05515-3.
- [23] Mark Pollicott and Richard Sharp. Exponential error terms for growth functions on negatively curved surfaces. *American Journal of Mathematics*, 120(5):1019–1042, 1998. doi: 10.1353/AJM.1998.0041.
- [24] David Ruelle. Zeta-functions for expanding maps and Anosov flows. *Inventiones Mathematicae*, 34(3):231–242, 1976. doi: 10.1007/BF01403069.
- [25] Atle Selberg. Harmonic analysis and discontinuous groups in weakly symmetric Riemannian spaces with applications to Dirichlet series. *Journal of the Indian Mathematical Society*, 20:47–87, 1956. URL <https://www.i-scholar.in/index.php/JIMSIMS/article/view/146884>.
- [26] Richard Sharp. Closed orbits in homology classes for Anosov flows. *Ergodic Theory and Dynamical Systems*, 13(2):387–408, 1993. doi: 10.1017/S0143385700007434.
- [27] Jan Snellman. *A Graded Subring of an Inverse Limit of Polynomial Rings*. PhD thesis, Stockholm University, 1998. URL <https://su.diva-portal.org/smash/record.jsf?pid=diva2%3A195258>.
- [28] Toshikazu Sunada. Riemannian coverings and isospectral manifolds. *Annals of Mathematics*, 121(1):169–186, 1985. doi: 10.2307/1971195.

- [29] A. B. Venkov and P. G. Zograf. On analogues of the artin factorization formulas in the spectral theory of automorphic functions connected with induced representations of fuchsian groups. *Mathematics of the USSR-Izvestiya*, 21(3):435–443, 1983. doi: 10.1070/IM1983v021n03ABEH001800. URL <https://www.mathnet.ru/eng/im1700>.
- [30] Ke Wang, Qiang Zhang, and Xuezhi Zhao. Word length formulae and normal forms of conjugacy classes in surface groups. arXiv preprint, 2025. URL <https://arxiv.org/abs/2511.12862>.